

OSSP_practical_5

2520030096_M Guru kiran

1.

```
guru_kiran@LAPTOP-OTG3R07B:~$ gedit prodscons.c
1#include<stdio.h>
2#include<unistd.h>
3#include<time.h>
4int main()
5{
6    int fd[2]; //declaring the pipe
7    pipe(fd); //creating the pipe
8    int pid; //declaring variable for data and pid
9    pid=fork(); //creating process
10    int data;
11    double start,end;
12    start=clock();
13    double timetaken;
14    if(pid==0){ //checking for child
15        close(fd[1]); //closing the write end of the pipe because child is reading
16        for(int i=0;i<5;i++){ //creating 5 consumers
17            read(fd[0],&data,sizeof(data)); //child is reading the data from the pipe
18            printf("Consumed %d\n",data); //printing the consumed data
19        }
20        close(fd[0]); //closing the read end of the data
21    }else{
22        printf("Producer\n");
23        close(fd[0]); //closing the read end of the pipe
24        for(int i=0;i<5;i++){ //printing 5 consumers
25            data=i*10;
26            write(fd[1],&data,sizeof(data));
27            printf("Produced %d\n",data);
28        }
29        close(fd[1]);
30        end=clock();
31        timetaken=(double)(end-start)/CLOCKS_PER_SEC; //calculating the communication time
32        printf("Communication time is %f",timetaken); //printing the time
33    }
34}
guru_kiran@LAPTOP-OTG3R07B:~$ gcc prodscons.c
guru_kiran@LAPTOP-OTG3R07B:~$ ./a.out
Parent process 2889
Child process 2890
Parent process completed
** (gedit:2935): WARNING **: 04:07:59.960: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:2935): WARNING **: 04:07:59.961: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
guru_kiran@LAPTOP-OTG3R07B:~$ gedit prodscons.c
** (gedit:3526): WARNING **: 04:08:07.787: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:3526): WARNING **: 04:08:07.787: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
```

```
guru_kiran@LAPTOP-OTG3R07B:~$ gedit prodscons.c
** (gedit:3526): WARNING **: 04:08:07.787: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:3526): WARNING **: 04:08:07.787: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
guru_kiran@LAPTOP-OTG3R07B:~$ gcc prodscons.c
guru_kiran@LAPTOP-OTG3R07B:~$ ./a.out
Producer
Produced 10
Produced 20
Produced 30
Produced 40
Produced 50
Consumer
Communication time is 0.000071Consumed 10
Consumed 20
Consumed 30
Consumed 40
Consumed 50
guru_kiran@LAPTOP-OTG3R07B:~$ |
```

2.

```
guru_kiran@LAPTOP-OTG3R0:~$ gedit waitex.c

** (gedit:873): WARNING **: 04:00:36.385: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:873): WARNING **: 04:00:36.385: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
^C
guru_kiran@LAPTOP-OTG3R0:~$ gedit program2.c

1#include<unistd.h>
2#include<stdio.h>
3int main()
4{
5    int fd[2];
6    pipe(fd);
7    int pid=fork();
8    if(pid==0)
9    {
10    close(fd[0]);
11    dup2(fd[1],1);
12    execlp("ls","ls","-l",NULL);
13    }
14    if(pid==0)
15    {
16    close(fd[1]);
17    dup2(fd[0],0);
18    execlp("grep","grep",".c",NULL);
19    }
20    }
21

Saving file ~/program2.c...

guru_kiran@LAPTOP-OTG3R0:~$ gedit program2.c

** (gedit:5778): WARNING **: 04:33:57.328: Could not load Peas repository: Typelib file for namespace 'Peas', version '1.0' not found
** (gedit:5778): WARNING **: 04:33:57.328: Could not load PeasGtk repository: Typelib file for namespace 'PeasGtk', version '1.0' not found
```

```
guru_kiran@LAPTOP-OTG3R0:~$ gcc program2.c
guru_kiran@LAPTOP-OTG3R0:~$ ./a.out
--W--Xr-t 1 guru_kiran guru_kiran 0 Jul 30 05:38 SUIIIIIIIIII.c
-rw-r--r-- 1 guru_kiran guru_kiran 204 Aug 31 08:25 Signal.c
-rw-r--r-- 1 guru_kiran guru_kiran 88 Jul 27 08:38 a.c
-rw-r--r-- 1 guru_kiran guru_kiran 198 Aug 11 05:41 allexec.c
-rw-r--r-- 1 guru_kiran guru_kiran 9 Aug 3 08:34 c2
-rw-r--r-- 1 guru_kiran guru_kiran 264 Sep 10 09:43 client.c
-rw-r--r-- 1 guru_kiran guru_kiran 71 Aug 11 04:54 excex1.c
-rw-r--r-- 1 guru_kiran guru_kiran 99 Aug 11 04:59 excex.c
-rw-r--r-- 1 guru_kiran guru_kiran 83 Aug 11 05:12 excv.c
-rw-r--r-- 1 guru_kiran guru_kiran 184 Aug 1 09:28 fork.c
-rw-r--r-- 1 guru_kiran guru_kiran 192 Aug 31 06:59 generateSignal.c
-rw-r--r-- 1 guru_kiran guru_kiran 232 Sep 15 05:53 malloc.c
-rw-r--r-- 1 guru_kiran guru_kiran 416 Aug 27 05:45 namedpipeFIFO1.c
-rw-r--r-- 1 guru_kiran guru_kiran 694 Aug 27 05:39 namedpipeex.c
-rw-r--r-- 1 guru_kiran guru_kiran 651 Aug 1 09:45 open2.c
-rw-r--r-- 1 guru_kiran guru_kiran 233 Aug 3 08:26 orw.c
-rw-r--r-- 1 guru_kiran guru_kiran 247 Aug 4 05:30 pipe1.c
-rw-r--r-- 1 guru_kiran guru_kiran 371 Aug 4 05:51 pipe2.c
-rw-r--r-- 1 guru_kiran guru_kiran 563 Aug 4 06:06 pipe3.c
-rw-r--r-- 1 guru_kiran guru_kiran 285 Aug 8 09:32 pipe6.c
-rw-r--r-- 1 guru_kiran guru_kiran 382 Aug 8 09:45 pipe7.c
-rw-r--r-- 1 guru_kiran guru_kiran 225 Aug 11 05:47 pipeex.c
-rw-r--r-- 1 guru_kiran guru_kiran 144 Aug 11 06:02 pipesort.c
-rw-r--r-- 1 guru_kiran guru_kiran 637 Aug 23 07:13 practical.c
-rw-r--r-- 1 guru_kiran guru_kiran 419 Aug 22 05:42 practical1.c
-rw-r--r-- 1 guru_kiran guru_kiran 550 Aug 22 09:42 practical2.c
-rw-r--r-- 1 guru_kiran guru_kiran 949 Sep 17 03:47 practical3.c
-rw-r--r-- 1 guru_kiran guru_kiran 998 Sep 17 04:08 prodscons.c
-rw-r--r-- 1 guru_kiran guru_kiran 239 Sep 17 04:34 program2.c
-rw-r--r-- 1 guru_kiran guru_kiran 6 Jul 30 10:13 public.txt
-rw-r--r-- 1 guru_kiran guru_kiran 87 Aug 3 06:48 read1.c
-rw-r--r-- 1 guru_kiran guru_kiran 235 Aug 3 08:00 readwrite.c
-rw-r--r-- 1 guru_kiran guru_kiran 216 Aug 3 08:41 removeb2.c
-rw-r--r-- 1 guru_kiran guru_kiran 0 Jul 30 10:12 script.sh
-rw-r--r-- 1 guru_kiran guru_kiran 243 Aug 27 04:56 segmentation.c
-rw-r--r-- 1 guru_kiran guru_kiran 258 Aug 23 08:46 state.c
-rw-r--r-- 1 guru_kiran guru_kiran 11447 Aug 22 09:57 trace.txt
-rw-r--r-- 1 guru_kiran guru_kiran 562 Sep 17 04:00 waitex.c
-rw-r--r-- 1 guru_kiran guru_kiran 0 Jul 30 05:57 z.c
-rw-r--r-- 1 guru_kiran guru_kiran 362 Sep 17 04:01 zombie.c
```